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Term work

of

Computer Networks -I Lab (PCS – 514)

Submitted in partial fulfilment of the requirement for the V semester

Bachelor of Technology

By

Aditya Pandey

2361030

Under the Guidance of

Ms. Vaishali Dev

Faculty In Charge

Department of

CSE

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

GRAPHIC ERA HILL UNIVERSITY, BHIMTAL CAMPUS

SATTAL ROAD, P.O.

BHOWALI DISTRICT- NAINITAL-

263132 2025-2026



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CERTIFICATE

The term work of Computer Networks Lab -I , being submitted by **Aditya Pandey** S/O **Hem Chandra Pandey** University Roll Number **2361030** to Graphic Era Hill University, Bhimtal Campus for the award of bonafide work carried out by her. She has worked under my guidance and supervision and fulfilled the requirement for the submission of this work report.

(Ms. Vaishali Dev)

Faculty In Charge

(Dr. Ankur Singh Bisht)

HOD, CSE Dept.



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Aditya Pandey
University Roll Number: 2361030



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PROBLEM STATEMENT :

1. Familiarization of Network Environment, Understanding and using network utilities: ipconfig, netstat, ping, telnet, ftp, traceroute etc.

SOLUTION :

1. ipconfig (Internet Protocol Configuration) ipconfig is a command-line utility available on Windows systems that displays and manages the IP address configuration of network interfaces. It is commonly used to view the current IP address, subnet mask, and default gateway assigned to the system. Advanced options allow users to release and renew IP addresses via DHCP. **Command Examples:**

```
ipconfig    # Shows basic network info
```

```
ipconfig /all      # Shows detailed info (MAC address, DNS, DHCP status etc.)
```

2. ifconfig (Interface Configuration) ifconfig is the Linux/macOS equivalent of ipconfig. It is used to display, configure, and manage network interfaces. It can show IP addresses, MAC addresses, and allow enabling or disabling of interfaces.

Command Examples:

```
ifconfig    # Show IP and MAC of interfaces
```

```
sudo ifconfig eth0 down # Disable a network interface
```

3. ping (Packet Internet Groper) ping is a diagnostic tool used to test the reachability of a host on an IP network. It sends ICMP echo request packets to the target host and measures the time taken for the responses. It helps determine whether the destination device is online and how fast it responds.

Command Example:

```
ping google.com    # Sends packets to Google's  
servers ping 192.168.1.1 # Tests connectivity  
with local router
```

4. tracert (Windows) / traceroute (Linux/macOS) This utility traces the route that packets take to reach a destination host. It lists all the intermediate routers the packet passes through, along with the time taken at each hop. This is useful for identifying network bottlenecks or failures. **Command Example:**

```
tracert google.com
```

5. netstat (Network Statistics) netstat is a command-line utility that displays active network connections, listening ports, and network protocol statistics. It helps users monitor open connections, detect suspicious activity, and troubleshoot network issues.

Command Examples:

```
netstat    # Shows active connections
```

```
netstat -a  # Shows all active ports and listening ports
```

6. telnet (Teletype Network Protocol) telnet is a network protocol and command-line tool used to establish a text-based communication session with a remote host using the TCP/IP protocol. It is often used to test connectivity to a specific port (like 80 for HTTP or 25 for SMTP), though it is now mostly replaced by more secure alternatives like SSH.

Command Example:

```
telnet google.com 80    # Test connection to port 80 (HTTP)
```

7. ftp (File Transfer Protocol) ftp is a standard network protocol used to transfer files between a client and a server over a TCP-based network. The ftp command-line tool allows users to upload and download files, authenticate with remote servers, and navigate remote directories.

Command Example:

```
ftp ftp.example.com
```

PROBLEM STATEMENT :

2. Familiarization with Transmission media and tools: Co-axial cable, UTP cable, Crimping tool, Connectors etc. Preparing the UTP cable for cross and direct connection using crimping tool.

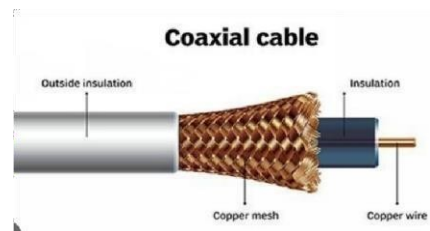
SOLUTION :

1. Coaxial Cable (Co-axial Cable)

A coaxial cable is a type of electrical cable with an inner conductor surrounded by a concentric conducting shield, separated by an insulating layer. It is commonly used for cable television, internet connections, and long-distance communication due to its excellent shielding from electromagnetic interference (EMI).

Use Case: Broadband internet, CCTV, cable TV

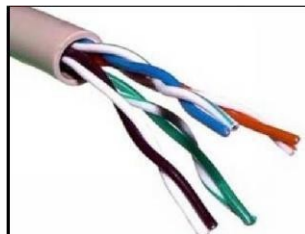
Connector Type: BNC, F-type



2. UTP Cable (Unshielded Twisted Pair)

A UTP cable consists of pairs of insulated copper wires twisted together. It lacks shielding, making it cheaper and more flexible, but more vulnerable to EMI. It is widely used in Ethernet networks and telephone systems. Types: Cat5, Cat5e, Cat6

Max Range (Ethernet): ~100 meters



3. Crimping Tool

A crimping tool is a hand tool used to attach connectors (such as RJ-45) to the ends of UTP cables. It ensures a secure and reliable electrical connection by pressing the connector's pins into the cable wires. Function: Terminating cables with RJ-45 connectors

4. Connectors (RJ-45)

RJ-45 connectors are modular plugs used to connect UTP cables to networking devices. They have 8 pins that correspond to the 8 wires in a UTP cable. These connectors are crucial in forming both straightthrough and crossover cables.

Pin Count: 8P8C (8 Positions, 8

Contacts) Use: Ethernet, LAN

connections



Preparing UTP Cable for Cross and Direct Connection

Straight-through Cable (Direct Connection)

Used to connect different types of devices (e.g., PC → Switch, PC → Router). Wiring Standard:

□ Both ends use same color coding (usually T568B) T568B Wiring Order:

1. Orange-White
2. Orange
3. Green-White
4. Blue
5. Blue-White
6. Green
7. Brown-White
8. Brown

Crossover Cable (Cross Connection)

Used to connect similar devices (e.g., PC ↔ PC, Switch ↔ Switch). Wiring Standard:

□ One end is T568A, the other is T568B.

T568A End:		T568B End:
1. Green-White	↔	Orange-White
2. Green	↔	Orange
3. Orange-White	↔	Green-White
4. Blue	↔	Blue
5. Blue-White	↔	Blue-White
6. Orange	↔	Green
7. Brown-White	↔	Brown-White
8. Brown	↔	Brown

Steps to Crimp a UTP Cable

1. Strip ~1 inch of the cable jacket using the crimping tool.
2. Untwist and align the wires as per the color code (T568A or B).
3. Trim wires evenly using the cutter.
4. Insert wires into the RJ-45 connector — ensure all reach the end.
5. Insert connector into the crimping slot and press hard.
6. Repeat on the other side as per the required cable type (cross/direct).

PROBLEM STATEMENT :

3. Installation and introduction of simulation tool. (Packet Tracer)

SOLUTION :

Introduction to Cisco Packet Tracer

Cisco Packet Tracer is a powerful network simulation and visualization tool developed by Cisco Systems. It is widely used in academic environments, particularly in Cisco Networking Academy programs, to provide students and networking professionals with a realistic and interactive platform for designing, configuring, and troubleshooting network topologies without requiring physical networking equipment. Packet Tracer supports a wide range of networking components, including routers, switches, wireless devices, PCs, and IoT devices. It provides both real-time and simulation modes, enabling users to observe packet flow and protocol behaviour in detail.

Key Features of Packet Tracer

Feature	Description
Device Simulation	Simulates routers, switches, PCs, laptops, servers, firewalls, and IoT devices.
Real-time Mode	Emulates live network behaviour for active packet flow and protocol execution.
Simulation Mode	Allows users to step through packet transmission for deeper analysis.
CLI Support	Provides a command-line interface similar to Cisco IOS for configuring devices.
Logical and Physical Views	Users can design topologies logically and visualize physical arrangements.
Multi-User Collaboration	Supports network collaboration in a classroom or remote learning environment

Importance of Packet Tracer

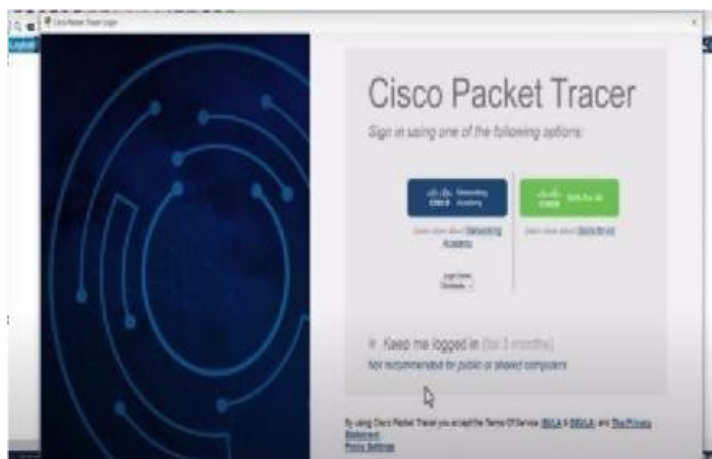
- Enables hands-on learning of networking concepts.
- Eliminates the cost of purchasing physical routers and switches.
- Ideal for practicing CCNA, CCNP, and other networking certifications.
- Facilitates experimentation and troubleshooting in a risk-free environment. □ Supports remote learning, making it accessible to students globally.

Installation Steps for Cisco Packet Tracer

Step-by-Step Installation Process (Windows)

1. Register on Cisco Networking Academy:
 - o Visit: <https://www.netacad.com>
 - o Create a free account and enrol in the *Introduction to Packet Tracer* course.
2. Download the Software:
 - o After enrolment, navigate to the Packet Tracer download section.
 - o Choose the appropriate version for your operating system.
3. Install Packet Tracer:
 - o Run the downloaded installer file.
 - o Accept the license agreement.
 - o Choose the installation directory (default is recommended).
 - o Complete the installation process.
4. Launch and Sign In:

- o Open Packet Tracer.
- o Login using your Cisco Networking Academy credentials



PROBLEM STATEMENT :

4. To configure a basic network topology consisting of routers, switches, and end devices such as PCs or laptops. Configure IP addresses and establish connectivity between devices.
(Using packet Tracer)

SOLUTION :

Step 1: Place Devices

- 1 Router (e.g 2811)
- 2 Switches (e.g 2960)
- 4 PCs (2 Per network i.e., PC0 & PC1 on SW1 , PC2 & PC3 on SW2)

Step 2: Connect with cables:

- PC0 -> SW1
- PC1 -> SW1
- PC2 -> SW2
- PC3 -> SW2
- SW1 -> Router
(Router's FastEthernet0/0)
- SW2 -> Router
(Router's FastEthernet0/1)

Step 3: Assign IP Addresses.

Network 1:

Router Fa0/0 -> 192.168.1.1/24

- PC0 -> 192.168.1.10/24 , Gateway : 192.168.1.1
- PC1 -> 192.168.1.11/24, Gateway : 192.168.1.1

Network 2:

Router Fa0/1 -> 192.168.2.1/24

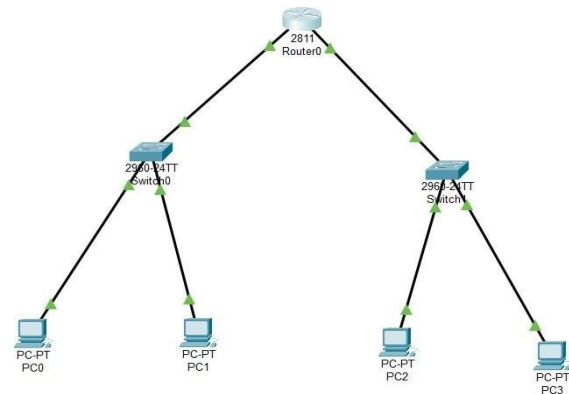
- PC2 -> 192.168.2.10/24, Gateway : 192.168.2.1
- PC3 -> 192.168.2.11/24, Gateway : 192.168.2.1

Step 4: Configure router

interfaces Click the Router -> CLI tab .

Type the following commands:

```
enable
configure terminal
hostname R1
interface FastEthernet0/0
ip address 192.168.1.1
255.255.255.0
no shutdown
interface FastEthernet0/1
ip address 192.168.2.1
255.255.255.0
```



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1(config-if)#interface FastEthernet0/1
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

R1(config-if)#exit
```

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	0.0.0.0
IPv6 Configuration	

no shutdown

exit

Step 5: Configure PCs

Click on each PC -> Desktop Tab -> IP Configuration.

PC0:

- IP: 192.168.1.10
- Subnet: 255.255.255.0
- Gateway: 192.168.1.1

Do the same for all PCs using IP s assigned.

Step 6: Test Connectivity.

Open PC0 -> Desktop -> Command Prompt .

Type:

ping 192.168.1.11 #PC 1
(same LAN)

ping 192.168.2.10 #PC2 (other LAN via router)

```
C:\>ping 192.168.1.11
```

```
Pinging 192.168.1.11 with 32 bytes of data:
```

```
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.1.11:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 192.168.2.10
```

```
Pinging 192.168.2.10 with 32 bytes of data:
```

```
Request timed out.
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
```

```
Ping statistics for 192.168.2.10:
```

```
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PROBLEM STATEMENT :

5. To configure a DHCP server on a router or a dedicated DHCP server device. Assign IP addresses dynamically to devices on the network and verify successful address assignment. (Using packet Tracer)

SOLUTION :

Step 1: Place Devices

- 1 Router (e.g 2811)
- 2 Switches (e.g 2960)
- 6 PCs (3 Per network i.e., PC0, PC1 & PC2 on SW1 ; PC3, PC4 & PC5 on SW2)

Step 2: Connect with cables:

- PC0 -> SW1 , PC1 -> SW1
- PC2 -> SW1
- PC3 -> SW2, PC4 -> SW2
- PC5 -> SW2
- SW1 -> Router (Router's FastEthernet0/0)
- SW2 -> Router (Router's FastEthernet0/1)

Step 3: Assign IP Addresses.

Network 1:

Router Fa0/0 -> 192.168.1.1/24

Network 2:

Router Fa0/1 -> 192.168.2.1/24

Other IP addresses assigned dynamically.

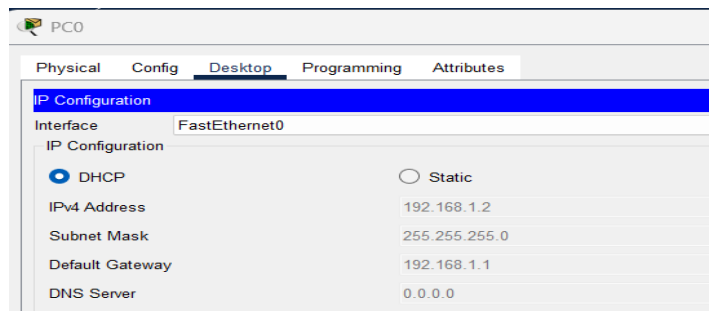
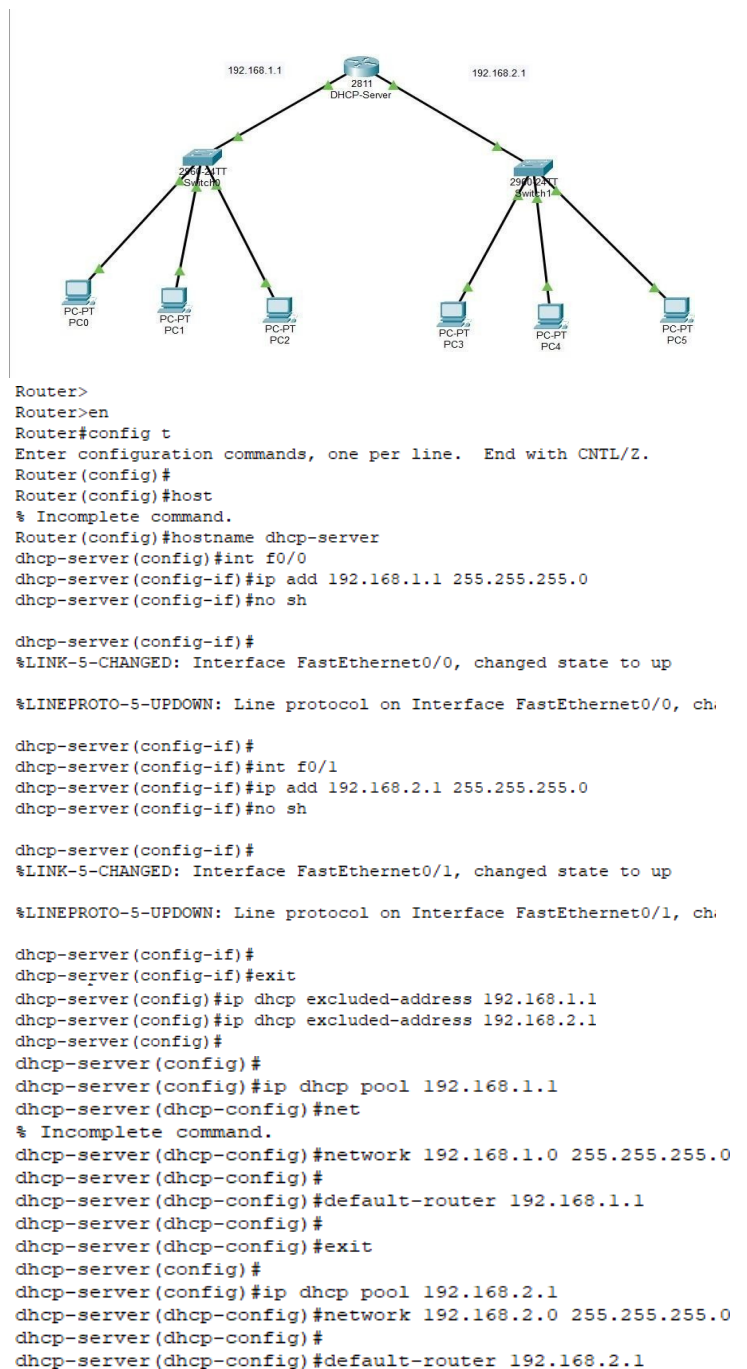
Step 4: Configure router

interfaces Click the Router -> CLI tab .

Type the following commands:

```
enable
configure terminal
hostname dhcp-server
interface FastEthernet0/0
ip address 192.168.1.1
255.255.255.0
no shutdown
interface FastEthernet0/1
ip address 192.168.2.1
255.255.255.0
no shutdown
exit
ip dhcp excluded-
address 192.168.1.1
ip dhcp excluded-address 192.168.2.1
ip dhcp pool 192.168.1.1
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
exit
ip dhcp pool 192.168.2.1
network 192.168.2.0 255.255.255.0
default-router 192.168.2.1
exit
```

Step 5: To verify , click on any PC -> Desktop -> IP Configuration -> DHCP .
You will see the IP address assigned to the respective PC.



PROBLEM STATEMENT :

6. To configure a local DNS server to resolve domain names within a network. (Using packet Tracer)

SOLUTION :

Step 1: Place Devices

- 1 Server
- 1 Switch (e.g 2960)
- 1 PC

Step 2: Connect with cables:

- PC0 -> Switch
- Switch -> Server

Step 3: Configure DNS Service on the server.

Click on the server -> Desktop -> IP

Configuration. Assign : IP Address: 192.168.1.1

Subnet Mask:

255.255.255.0 DNS Server :

192.168.1.1

Step 4: Host a HTTP service on this server.

Go to Services -> HTTP.

Turn on HTTP.

Click on (edit) next to index.html.

Edit the html file as below.

```
<html>
<h1> Hello Cisco </h1>
</html>
```

Click Save and choose Yes on the pop-up that appears.

Step 5: Set up DNS Service.

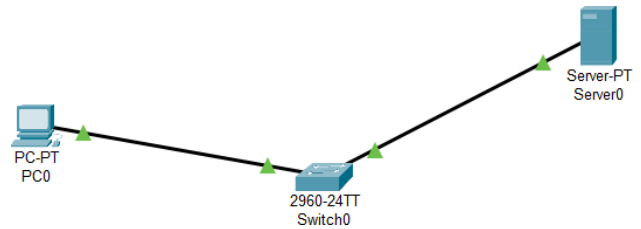
Go to DNS.

Turn DNS Service on.

Type a name in Name box. (e.g., mywebsite)

Type address as 192.168.1.1 (i.e. IP address of DNS Server).

Click Add.



IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.1.1
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	192.168.1.1

SERVICES	
☒ HTTP	☐ HTTPS
☐ DHCP	☐ Off
☐ DHCPv6	☐ On
☐ TFTP	
☐ DNS	
☐ SYSLOG	
☐ AAA	
☐ NTP	
☐ EMAIL	
☐ FTP	
☐ IoT	
☐ VM Management	
☐ Radius EAP	

File Manager	
File Name	Edit
1 copyrights.html	(edit)
2 cscoptlogo177x111.jpg	
3 helloworld.html	(edit)
4 image.html	(edit)
5 index.html	(edit)

DNS	
DNS Service	☒ On ☐ Off
Resource Records	
Name	mywebsite
Type	A Record
Address	192.168.1.1

No.	Name	Type	Address
0	mywebsite	A Record	192.168.1.1

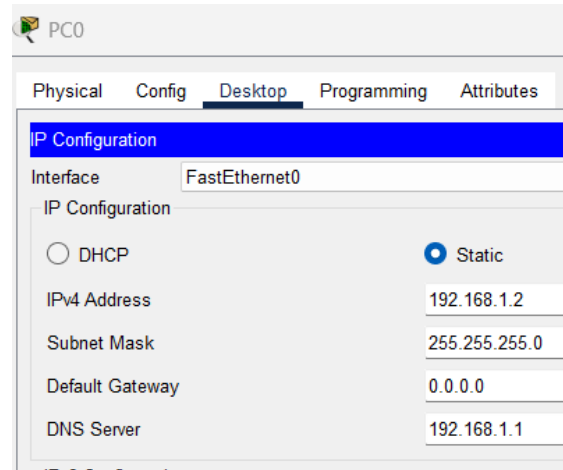
Step 6: Configure PC.

Click on PC0 -> Desktop -> IP Configuration. Update IP Address and DNS Server as given below:

IP Address: 192.168.1.2

Subnet Mask: 255.255.255.0

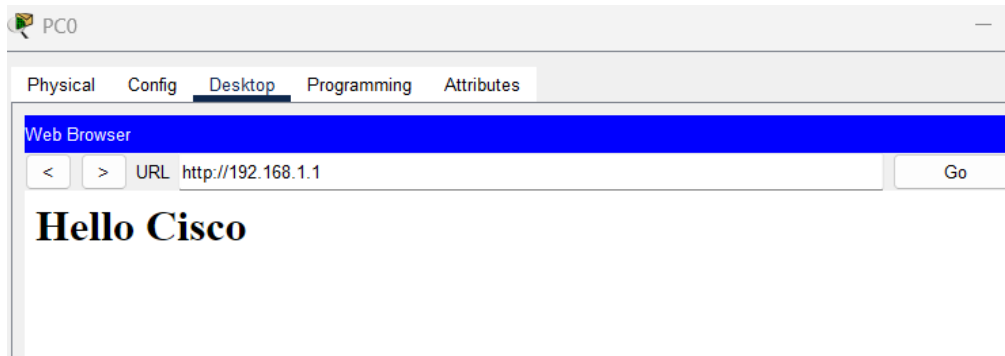
DNS Server: 192.168.1.1

**Step 7: Check DNS Configuration.**

Click on PC0 -> Desktop -> Web Browser.

In the URL box, type the name of the website (here, mywebsite) or IP Address (192.168.1.1).

If the DNS is configured, it will display the exact text you edited in the HTML file in the web browser.



PROBLEM STATEMENT :

7. To analyse complete TCP/IP protocol suite layer's headers using Wire Shark

SOLUTION :

Step 1: Install & Open Wireshark

- Download and install the latest version of Wireshark.
- Ensure **Npcap** is installed.
- Open Wireshark as Administrator.
- Select your active network interface (WiFi/Ethernet).

Step 2: Start Packet Capture

- Click **Start Capturing Packets** (Blue fin icon).
- Generate traffic by:
 - Opening a website.
 - Refreshing a page
 - Using commands like **ping google.com**

Step 3: Apply protocol filters.

Use the following filters to view headers of each layer: Application Layer Filters

Protocol

Filter DNS

dns

HTTP http

HTTP/2http2

Transport Layer Filters

Protocol

Filter TCP

tcp

UDP

udp Network Layer

Filters

Protocol

Filter IPv4

ip

IPv6

ipv6 Data Link Layer

Filters

Protocol

Filter

Ethernet

eth

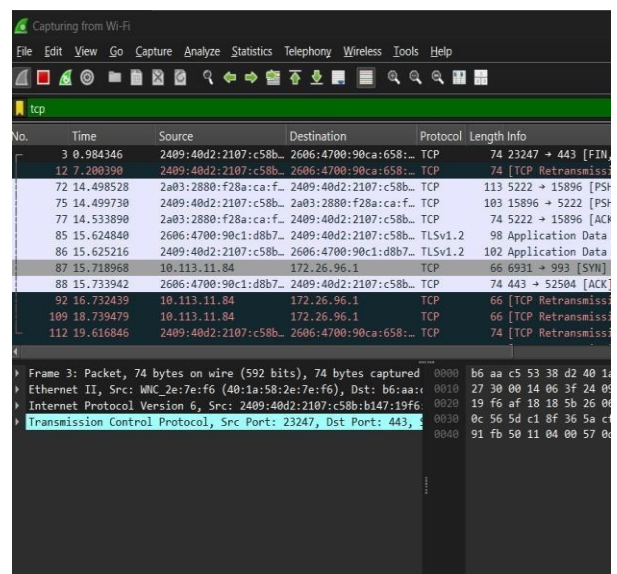
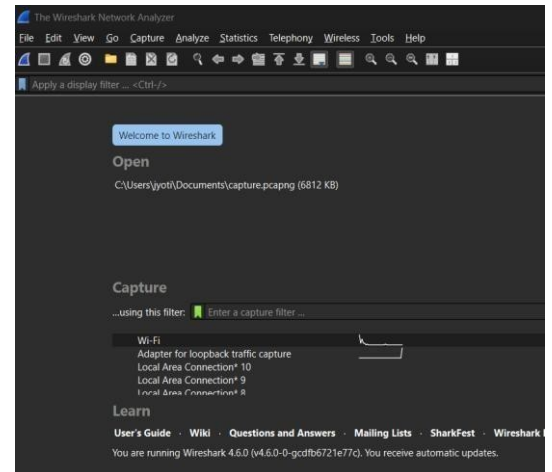
Example: To track packets to/from a specific server (Google):

ip.addr == 142.250.183.206

Step 4: Analyse packet details

Click any captured packet to analyse all headers:

- Ethernet Header
- IP Header
- TCP/UDP Header
- Application Layer Data



Current filter: ip.addr == 142.250.183.206				
No.	Time	Source	Destination	Protocol
3912	135.404122	192.168.1.97	142.250.183.206	TCP
3913	135.404327	192.168.1.97	142.250.183.206	TCP
3914	135.420522	192.168.1.97	142.250.183.206	TCP
3920	135.463227	142.250.183.206	192.168.1.97	TCP
3921	135.463333	192.168.1.97	142.250.183.206	TCP
3928	135.482145	142.250.183.206	192.168.1.97	TCP
3929	135.482275	192.168.1.97	142.250.183.206	TCP
3930	135.489796	142.250.183.206	192.168.1.97	TCP
3931	135.489883	192.168.1.97	142.250.183.206	TCP
3932	135.490338	192.168.1.97	142.250.183.206	TCP
3933	135.490338	192.168.1.97	142.250.183.206	TLSv1.3
3934	135.637353	192.168.1.97	142.250.183.206	TCP
3954	135.667925	142.250.183.206	192.168.1.97	TCP
3956	135.698824	142.250.183.206	192.168.1.97	TCP
3957	135.698893	192.168.1.97	142.250.183.206	TCP
3958	135.699068	142.250.183.206	192.168.1.97	TLSv1.3
3959	135.699068	142.250.183.206	192.168.1.97	TCP
3961	135.699151	192.168.1.97	142.250.183.206	TCP
3962	135.699169	192.168.1.97	142.250.183.206	TCP
3964	135.699219	142.250.183.206	192.168.1.97	TCP
3965	135.699989	192.168.1.97	142.250.183.206	TLSv1.3
3966	135.701406	192.168.1.97	142.250.183.206	TCP
3967	135.708065	192.168.1.97	142.250.183.206	HTTP
3984	135.835251	142.250.183.206	192.168.1.97	TCP
3985	135.835251	142.250.183.206	192.168.1.97	TCP
3987	135.835251	142.250.183.206	192.168.1.97	TCP
3988	135.835251	142.250.183.206	192.168.1.97	TCP
3989	135.835404	192.168.1.97	142.250.183.206	TCP
3990	135.839722	142.250.183.206	192.168.1.97	TCP

The fields will appear in the detailed decode pane.

Step 5: End Capture

Click Stop .

Save the capture file if needed.

PROBLEM STATEMENT :

8. Static Routing: Configure static routes on multiple routers to enable communication between different networks. Test the connectivity by pinging between hosts in different networks.
(Using packet Tracer)

SOLUTION :

Step 1: Place Devices

- 2 Router (e.g 2811)
- 2 Switches (1 on each router)
- 4 PCs (2 Per network i.e., PC0 & PC1 on SW1 , PC2 & PC3 on SW2)

Step 2: Connect with cables:

- PC0 -> SW1 , PC1 -> SW1
- PC2 -> SW2 , PC3 -> SW2
- SW1 -> Router 0 (Router's FastEthernet0/0)
- SW2 -> Router 1 (Router's FastEthernet0/0)
- Router 0 -> Router 1
(Router's FastEthernet0/1)

Step 3: Assign IP

Addresses. Network 1:

Router 0 Fa0/0 -> 192.168.10.1/24

- PC0 -> 192.168.10.10/24 , Gateway : 192.168.10.1
- PC1 -> 192.168.10.20/24, Gateway : 192.168.10.1

Network 2:

Router 1 Fa0/0 -> 192.168.20.1/24

- PC2 -> 192.168.20.10/24, Gateway : 192.168.20.1
- PC3 -> 192.168.20.20/24, Gateway : 192.168.20.1

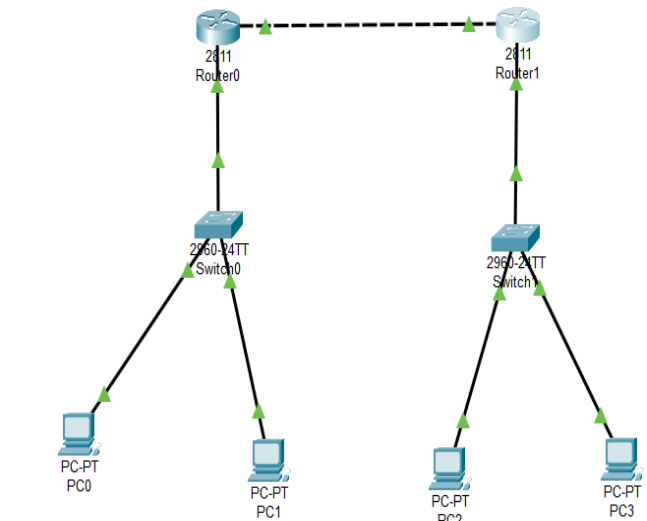
Network 3:

Router 0 Fa0/1 ->
10.0.12.1 Router 1
Fa0/1 ->

10.0.12.2 **Step 4:** Configure router
interfaces Click the Router0 -> CLI
tab .

Type the following commands:

```
enable
configure terminal
interface f0/0
ip add 192.168.10.1 255.255.255.0
no sh
exit
int f0/1
ip add 10.0.12.1 255.255.255.252
no sh
exit
ip route 192.168.20.0 255.255.255.0
10.0.12.2
end
write memory
```



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface
Processor board ID JAD05190MT2 (42923591495)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)
Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface f0/0
Router(config-if)#ip add 192.168.10.1 255.255.255.0
Router(config-if)#no sh
Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int f0/1
Router(config-if)#ip add 10.0.12.1 255.255.255.252
Router(config-if)#no sh
Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#ip route 192.168.20.0 255.255.255.0
% Incomplete command.
Router(config)#ip route 192.168.20.0 255.255.255.0 10.0.12.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
```

Click the Router1 -> CLI tab .

Type the following

commands:

```
enable
configure terminal
interface f0/0
ip add 192.168.20.1 255.255.255.0
no sh
exit
int f0/1
ip add 10.0.12.2 255.255.255.252
no sh
exit
ip route 192.168.10.0 255.255.255.0 10.0.12.1
end
write memory
```



```
Router1
Physical Config CLI Attributes
IOS Command Line Interface
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)
Press RETURN to get started!
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int f0/0
Router(config-if)#ip add 192.168.20.1 255.255.255.0
Router(config-if)#no sh
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#int f0/1
Router(config-if)#ip add 10.0.12.2 255.255.255.252
Router(config-if)#no sh
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#ip route 192.168.10.0 255.255.255.0 10.0.12.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#write memory
Building configuration...
[OK]
```

Step 5: Configure PCs

Click on each PC -> Desktop Tab -> IP Configuration.

PC0:

- IP: 192.168.10.10
- Subnet: 255.255.255.0
- Gateway: 192.168.10.1

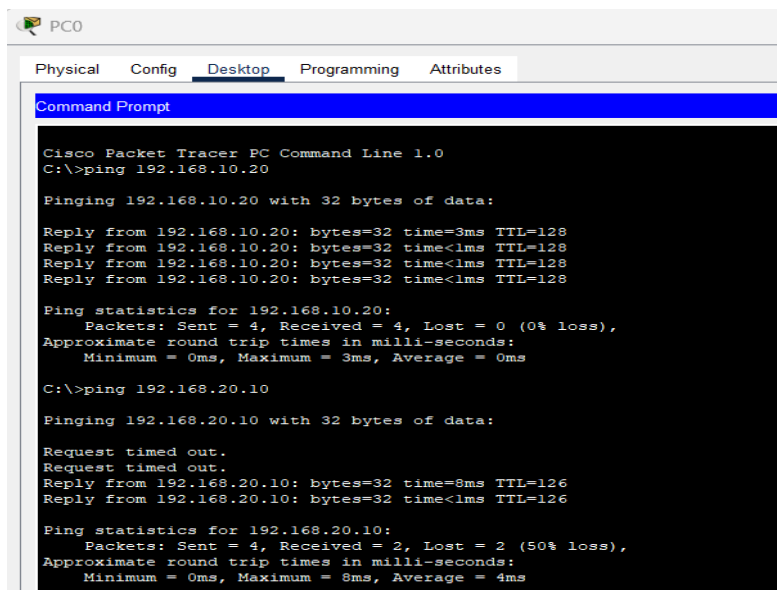
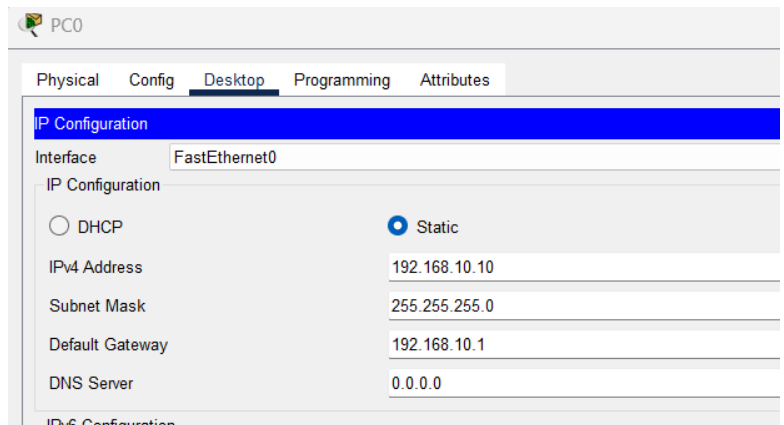
Do the same for all PCs using IP s assigned.

Step 6: Test Connectivity.

Open PC0 ->Desktop -> Command Prompt .

Type:

```
ping 192.168.10.20 #PC 1 (same LAN)
ping 192.168.20.10 #PC2 (other LAN via router)
```



PROBLEM STATEMENT :

9. Dynamic Routing (RIP): Configure routers to use the Routing Information Protocol (RIP) for dynamic routing. Enable RIP on the interfaces connected to different networks and verify that routes are being learned and propagated. Test the connectivity between hosts in different networks.

SOLUTION :

Step 1: Place Devices

- 2 Router (e.g 2811)
- 2 Switches (1 on each router)
- 4 PCs (2 Per network i.e., PC0 & PC1 on SW1 , PC2 & PC3 on SW2)

Step 2: Connect with cables:

- PC0 -> SW1 , PC1 -> SW1
- PC2 -> SW2 , PC3 -> SW2
- SW1 -> Router 0 (Router's FastEthernet0/0)
- SW2 -> Router 1 (Router's FastEthernet0/0)
- Router 0 -> Router 1 (Router's FastEthernet0/1)

Step 3: Assign IP

Addresses. Network 1:

Router 0 Fa0/0 -> 192.168.1.1/24

- PC0 -> 192.168.1.10/24 , Gateway : 192.168.1.1
- PC1 -> 192.168.1.20/24, Gateway : 192.168.1.1

Network 2:

Router 1 Fa0/0 -> 192.168.2.1/24

- PC2 -> 192.168.2.10/24, Gateway : 192.168.2.1
- PC3 -> 192.168.2.20/24, Gateway : 192.168.2.1

Network 3:

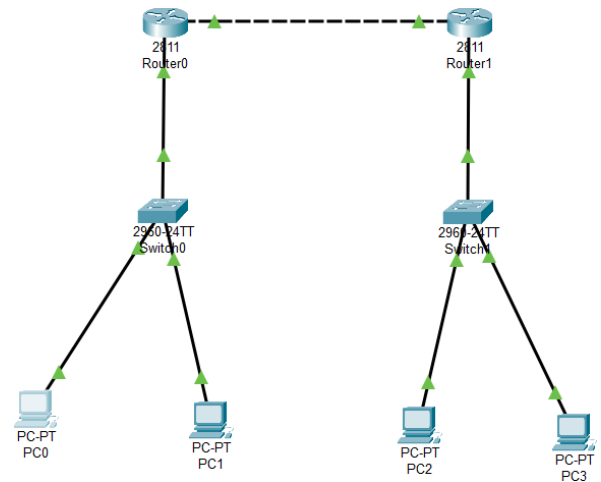
Router 0 Fa0/1 -> 10.0.0.1

Router 1 Fa0/1 -> 10.0.0.2

Step 4: Configure router interfaces Click the Router0 -> CLI tab .

Type the following commands:

```
enable
configure terminal
hostname R1
interface f0/0
ip add 192.168.1.1
255.255.255.0
no sh
exit
int f0/1
ip add 10.0.0.1
255.255.255.0
no sh
exit
router rip
```



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface
Would you like to enter the initial configuration dialog? [yes/no]: no
Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface f0/0
R1(config-if)#ip add 192.168.1.1 255.255.255.0
R1(config-if)#no sh

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1(config-if)#exit
R1(config)#int f0/1
R1(config-if)#ip add 10.0.0.1 255.255.255.0
R1(config-if)#no sh

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

R1(config-if)#exit
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#network 192.168.1.0
R1(config-router)#network 10.0.0.0
R1(config-router)#no auto-summary
R1(config-router)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#write memory
Building configuration...
[OK]
```

```

version 2
network 192.168.1.0
network 10.0.0.0
no auto-summary
end

```

write memory Click
the Router1 -> CLI tab.

Type the following

commands:

```

enable
configure terminal
hostname R2
interface f0/0
ip add 192.168.2.1
    255.255.255.0
no sh
exit
int f0/1
ip add 10.0.0.2 255.255.255.0
no sh
exit
router rip
version 2
network 192.168.2.0
network 10.0.0.0
no auto-summary
end
write memory

```

Step 5: Configure PCs

Click on each PC -> Desktop Tab -> IP Configuration.

PC0:

- IP: 192.168.1.10
- Subnet: 255.255.255.0
- Gateway: 192.168.1.1

Do the same for all PCs using IP
s assigned.

Step 6: Test Connectivity.

Open PC0 -> Desktop -> Command
Prompt.

Type:

```

ping 192.168.1.20 #PC 1
    (same LAN)
ping 192.168.2.10 #PC2
    (other LAN via router)

```

```

Router1
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#interface f0/0
R2(config-if)#ip add 192.168.2.1 255.255.255.0
R2(config-if)#no sh

R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R2(config-if)#exit
R2(config)#interface f0/1
R2(config-if)#ip add 10.0.0.2 255.255.255.0
R2(config-if)#no sh

R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

R2(config-if)#exit
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#network 192.168.2.0
R2(config-router)#network 10.0.0.0
R2(config-router)#no auto-summary
R2(config-router)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#write memory
Building configuration...
[OK]

```

PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.10

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

IPv6 Configuration

```

PC0
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

PROBLEM STATEMENT :

10. Dynamic Routing (OSPF): Configure routers to use the Open Shortest Path First (OSPF) routing protocol. Set up OSPF on the routers and advertise network information. Verify that OSPF is establishing neighbour relationships and propagating routes. Test connectivity between hosts in different networks. (Using packet Tracer)

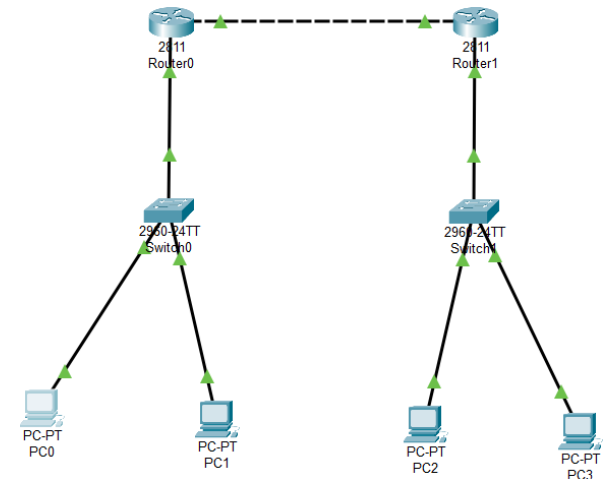
SOLUTION :

Step 1: Place Devices

- 2 Router (e.g 2811)
- 2 Switches (1 on each router)
- 4 PCs (2 Per network i.e., PC0 & PC1 on SW1 , PC2 & PC3 on SW2)

Step 2: Connect with cables:

- PC0 -> SW1 , PC1 -> SW1
- PC2 -> SW2 , PC3 -> SW2
- SW1 -> Router 0 (Router's FastEthernet0/0)
- SW2 -> Router 1 (Router's FastEthernet0/0)
- Router 0 -> Router 1 (Router's FastEthernet0/1)



Step 3: Assign IP Addresses.

Network 1:

Router 0 Fa0/0 -> 192.168.1.1/24

- PC0 -> 192.168.1.10/24 , Gateway : 192.168.1.1
- PC1 -> 192.168.1.20/24, Gateway : 192.168.1.1

Network 2:

Router 1 Fa0/0 -> 192.168.2.1/24

- PC2 -> 192.168.2.10/24, Gateway : 192.168.2.1
- PC3 -> 192.168.2.20/24, Gateway : 192.168.2.1

Network 3:

Router 0 Fa0/1 -> 192.168.3.1

Router 1 Fa0/1 -> 192.168.3.2

Step 4: Configure router

interfaces Click the Router0 -> CLI tab .

Type the following commands:

```
enable
configure terminal
hostname R1
interface f0/0
ip add 192.168.1.1
255.255.255.0
no sh
exit
int f0/1
ip add 192.168.3.1
255.255.255.0
no sh
```

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface f0/0
R1(config-if)#ip add 192.168.1.1 255.255.255.0
R1(config-if)#no sh

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1(config-if)#exit
R1(config)#int f0/1
R1(config-if)#ip add 192.168.3.1 255.255.255.0
R1(config-if)#no sh

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

R1(config-if)#exit
R1(config)#router ospf 1
R1(config-router)#network 192.168.1.0 0.0.0.255 area 0
R1(config-router)#network 192.168.3.0 0.0.0.255 area 0
R1(config-router)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#write memory
Building configuration...
[OK]
```

```

exit
router ospf 1
network 192.168.1.0 0.0.0.255
area 0
network 192.168.3.0 0.0.0.255
area 0
end

```

write memory Click

the Router1 -> CLI tab .

Type the following

commands:

```

enable
configure terminal
hostname R2
interface f0/0
ip add 192.168.2.1 255.255.255.0
no sh
exit
int f0/1
ip add 192.168.3.2
255.255.255.0
no sh
exit
router ospf 2
network 192.168.2.0 0.0.0.255
area 0
network 192.168.3.0
0.0.0.255 area 0
end
write memory

```

Step 5: Configure PCs

Click on each PC -> Desktop Tab -> IP Configuration.

PC0:

- IP: 192.168.1.10
- Subnet: 255.255.255.0
- Gateway: 192.168.1.1

Do the same for all PCs using IP s assigned. **Step 6:** Verify OSPF neighbor relationships and propagating routes.

```

ip ospf neighbor
show ip route

```

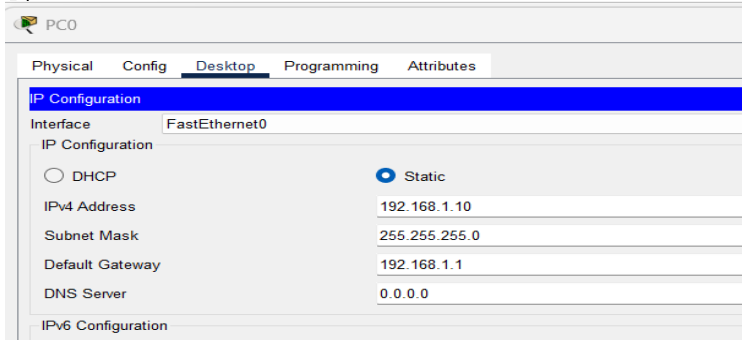
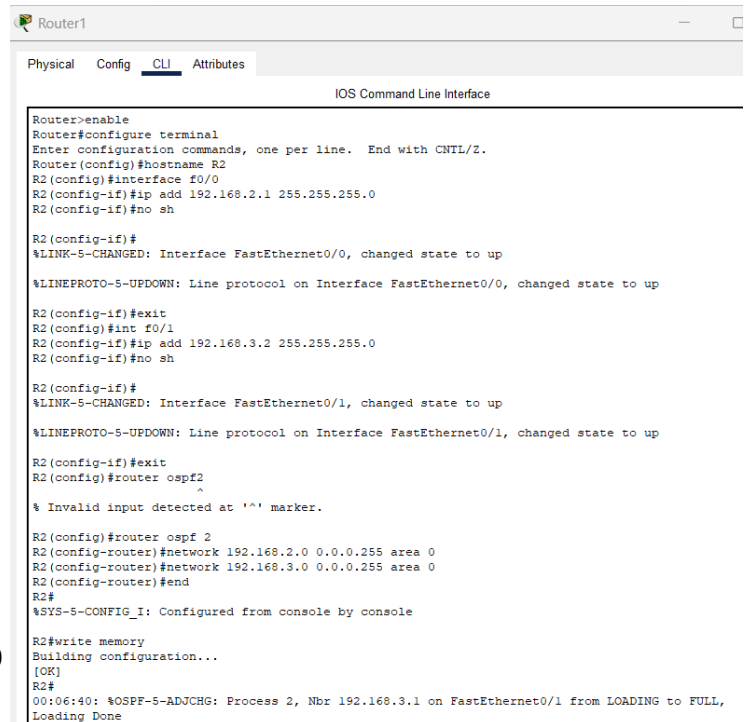
Step 7: Test Connectivity.

Open PC0 -> Desktop -> Command Prompt .
Type:

```

ping 192.168.1.20 #PC 1
(same LAN)
ping 192.168.2.10 #PC2
(other LAN via router)

```



```

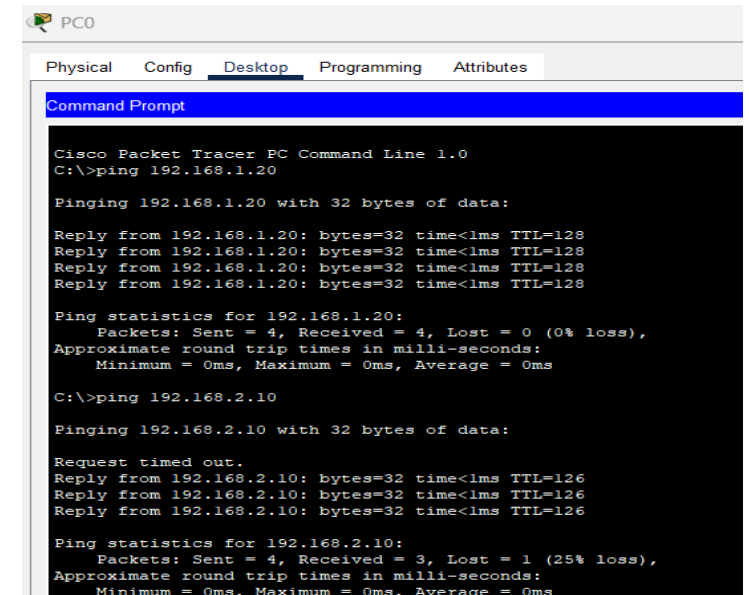
R1>show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
192.168.3.2 1 2WAY/DROTHER 00:00:32 192.168.3.2 FastEthernet0/1
R1>
00:00:40: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.2 on FastEthernet0/1 from LOADING to FULL, Loading Done

R1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.1.0/24 is directly connected, FastEthernet0/0
L 192.168.1.1/32 is directly connected, FastEthernet0/0
O 192.168.2.0/24 [110/2] via 192.168.3.2, 00:00:23, FastEthernet0/1
L 192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.3.0/24 is directly connected, FastEthernet0/1
L 192.168.3.1/32 is directly connected, FastEthernet0/1

```



PROBLEM STATEMENT :

11. TCP Client-Server Communication:

Implement a TCP client program that sends a message to a TCP server program. Implement the corresponding TCP server program that receives the message and displays it. Test the communication between the client and server by exchanging messages (Using 'C' Language)

SOLUTION :

1. TCP Server Code (C Language)

```
// File:
tcp_server.c
#include <stdio.h>
#include
<stdlib.h>
#include
<string.h>
#include <unistd.h>      // for close()
#include <netinet/in.h>  // for
sockaddr_in #define PORT 8080
#define BUFFER_SIZE
1024 int main() {
    int server_fd, new_socket;
    char buffer[BUFFER_SIZE] =
    {0}; struct sockaddr_in
    address; int addrlen =
    sizeof(address);
    // 1. Create socket
    server_fd = socket(AF_INET, SOCK_STREAM,
    0); if (server_fd == 0) {
        perror("Socket failed");
        exit(EXIT_FAILURE);
    }
    // 2. Bindsocket to IP/Port
    address.sin_family = AF_INET;
    address.sin_addr.s_addr = INADDR_ANY; // Accept connections from any IP
    address.sin_port = htons(PORT);
    if (bind(server_fd, (struct sockaddr*)&address, sizeof(address)) < 0)
        { perror("Bind failed");
        exit(EXIT_FAILURE);
    }
    // 3. Listen
    if (listen(server_fd, 3) < 0)
        { perror("Listen failed");
        exit(EXIT_FAILURE);
    }
    printf("Server is listening on port %d...\n", PORT);
    // 4. Accept connection
    new_socket = accept(server_fd, (struct sockaddr*)&address,
    (socklen_t*)&addrlen);
    if (new_socket < 0) {
        perror("Accept failed");
        exit(EXIT_FAILURE);
    }
    // 5. Read message from client
    read(new_socket, buffer, BUFFER_SIZE);
    printf("Message from client: %s\n",
    buffer);
    // Optional: Send a reply
    char *reply = "Hello from server!";
    send(new_socket, reply, strlen(reply), 0);
```

```

// 6. Close sockets
close(new_socket); close(server_fd); return 0;
}

```

2. TCP Client Code (C Language)

```

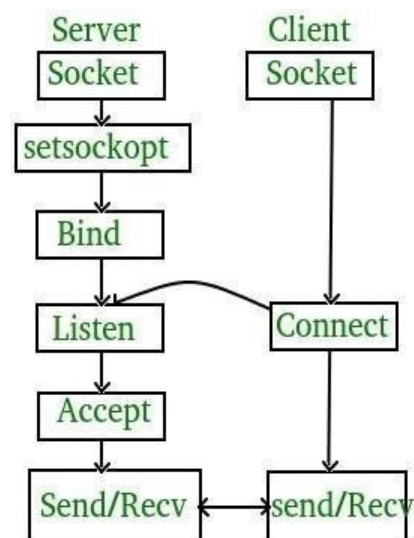
//File:tcp_client.c
#include<stdio.h>
#include<stdlib.h>
#include<string.h>
#include<unistd.h>
//for close()
#include<arpa/inet.h> //for sockaddr_in, inet_addr
#define PORT 8080
#define BUFFER_SIZE
1024 int main() {
    int sock = 0;
    struct sockaddr_in serv_addr;
    char *message = "Hello from
client!"; char buffer[BUFFER_SIZE]
= {0};
    // 1. Create socket
    sock = socket(AF_INET, SOCK_STREAM,
0); if (sock < 0) {
        printf("\n Socket creation error
\n"); return -1;
    }
    // 2. Set server address
    serv_addr.sin_family = AF_INET;
    serv_addr.sin_port = htons(PORT);
    // Convert IPv4 addresses from text to binary
    if (inet_pton(AF_INET, "127.0.0.1", &serv_addr.sin_addr)
    <= 0) { printf("\nInvalid address/Address not
supported \n"); return -1;
    }
    // 3. Connect to server
    if (connect(sock, (struct sockaddr*)&serv_addr, sizeof(serv_addr)) < 0)
    { printf("\nConnection Failed \n");
    return -1;
    }
    // 4. Send message to server
    send(sock, message, strlen(message), 0);
    printf("Message sent to server: %s\n",
message);
    // 5. Read server reply
    read(sock, buffer,
BUFFER_SIZE);
    printf("Message from server:
%s\n", buffer);
    // 6. Close
    socket
    close(sock);
    return 0;
}

```

OUTPUT

:

Server (Terminal 1):
 Server is listening on port 8080...
 Message from client: Hello from client!
 Client (Terminal 2):
 Message sent to server: Hello from client!
 Message from server: Hello from server!



PROBLEM STATEMENT :

12. UDP Client-Server Communication : Implement a UDP Client program that sends a message to UDP server program. Implement the corresponding UDP server program that receives the message and displays it (Using 'C' Language)

SOLUTION :

1: UDP Server Code

```
(udp_server.c) #include
<stdio.h> #include
<stdlib.h> #include
<unistd.h> #include
<string.h> #include
<arpa/inet.h> #define
PORT 8080
#define BUFFER_SIZE
1024 int main() {
    int sockfd;
    char buffer[BUFFER_SIZE];
    struct sockaddr_in servaddr, cliaddr;
    // 1. Create socket
    sockfd = socket(AF_INET, SOCK_DGRAM,
0); if (sockfd < 0) {
        perror("Socket creation failed");
        exit(EXIT_FAILURE);
    }
    // 2. Define server address
    memset(&servaddr, 0, sizeof(servaddr));
    memset(&cliaddr, 0, sizeof(cliaddr));
    servaddr.sin_family = AF_INET;
    // IPV4
    servaddr.sin_addr.s_addr = INADDR_ANY; // Any
    IP servaddr.sin_port = htons(PORT);
    // Port
    // 3. Bind the socket with the server address
    if (bind(sockfd, (const struct sockaddr *)&servaddr, sizeof(servaddr))
< 0)
    {
        perror("Bind
        failed");
        close(sockfd);
        exit(EXIT_FAILURE);
    }
    printf("UDP Server is running on port %d...\n", PORT);
    // 4. Receive message
    int len, n;
    len = sizeof(cliaddr);
    n = recvfrom(sockfd, buffer, BUFFER_SIZE, 0, (struct sockaddr
    *) &cliaddr, &len);
    buffer[n] = '\0';
    // 5. Display message
    printf("Message from client: %s\n", buffer);
    // Optional: Send reply
    char *reply = "Message received!";
    sendto(sockfd, reply, strlen(reply), 0, (struct sockaddr *)
    &cliaddr, len);
    // 6. Close socket
```

```

        close(sockfd
    ); return 0;
}
2: UDP Client Code (udp_client.c)
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <arpa/inet.h>
#define PORT 8080
#define BUFFER_SIZE
1024 int main() {
    int sockfd;
    char buffer[BUFFER_SIZE];
    char *message = "Hello from
client!"; struct sockaddr_in
servaddr;
    // 1. Create socket
    sockfd = socket(AF_INET, SOCK_DGRAM,
0); if (sockfd < 0) {
        perror("Socket creation failed");
        exit(EXIT_FAILURE);
    }
    // 2. Define server address
    memset(&servaddr, 0, sizeof(servaddr));
    servaddr.sin_family = AF_INET;
    servaddr.sin_port = htons(PORT);
    servaddr.sin_addr.s_addr = inet_addr("127.0.0.1");
    // 3. Send message to server
    sendto(sockfd, message, strlen(message), 0, (const struct sockaddr
*) &servaddr,
        sizeof(servaddr)
    );
    printf("Message sent to server: %s\n", message);
    // 4. Receive reply from
    server int len, n;
    len = sizeof(servaddr);
    n =recvfrom(sockfd, buffer, BUFFER_SIZE, 0, (struct sockaddr *)
&servaddr, &len);
    buffer[n] = '\0';
    printf("Message from server: %s\n", buffer);
    // 5. Close
    socket
    close(sockfd);
    return 0;}

```

OUTPUT :

Server Output:

UDP Server is running on port 8080...

Message from client: Hello from client!

Client Output:

Message sent to server: Hello from client!

Message from server: Message received!

